

Logistics Module Construction & Existing FS Integration Workflow

Engineering Methodology: Building the Logistics Engine from Scratch while Integrating into the Operational Financial Statements (FS) System

TARGET COMPLETION	CORE ARCHITECTURE	EXISTING ENGINE	STATUS
December 2026	Bridge-First / Contract-Driven	FS Accounting (Live & Verified)	Approved Strategy

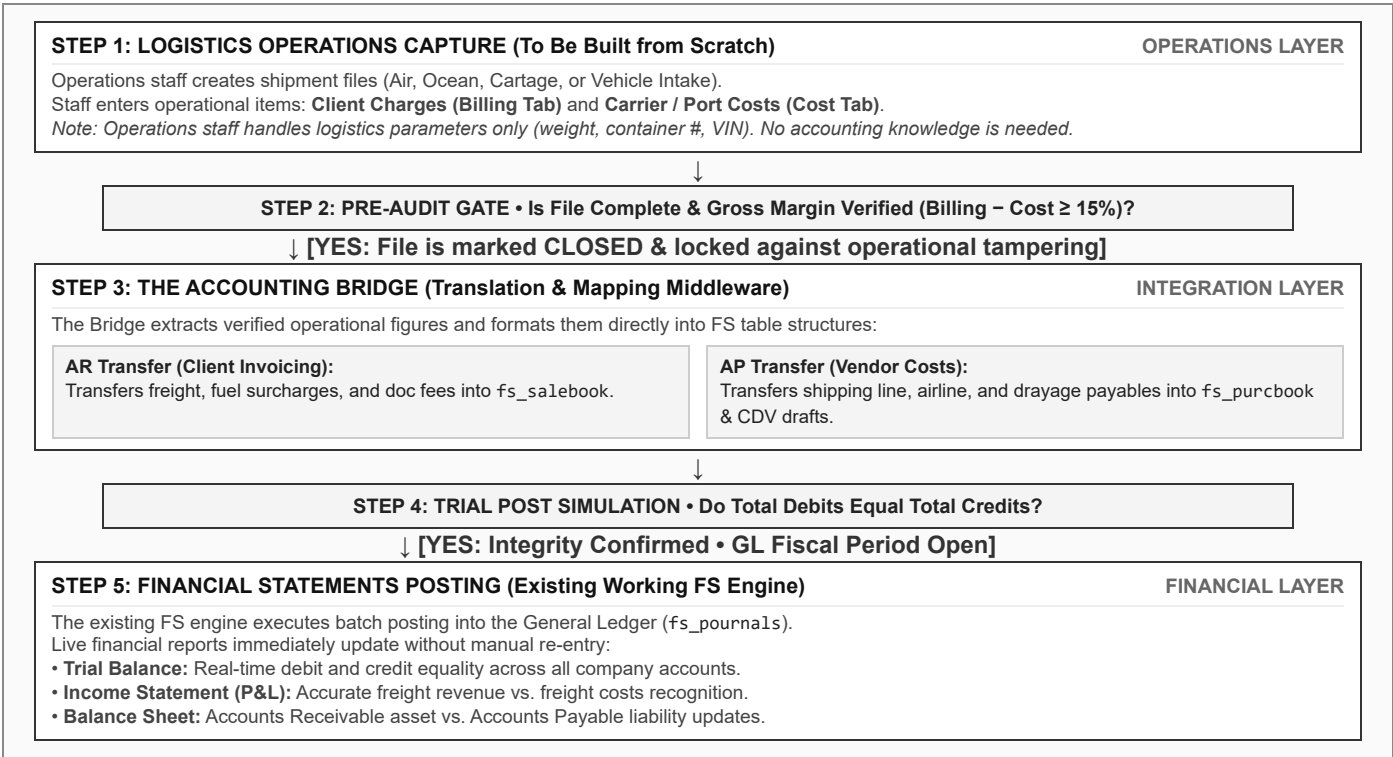
1. EXECUTIVE ANSWER TO MANAGEMENT: THE STRATEGIC APPROACH

The core management inquiry asks: "How are we going to make the logistics app that integrates with the FS that is already working? Are you making the logistics part first then putting FS later?"

Direct Answer: We do **not** rebuild or alter the working FS system later. Instead, we use a **Bridge-First / Contract-Driven Engineering Strategy**: The working Financial Statements (FS) system is already proven, so its database schema acts as our fixed, unchanging target. We build the unbuilt Logistics engine *specifically designed to output into that exact FS schema* via an automated Translation Bridge.

- The FS Interface Contract is Frozen First:** We specify the exact fields the working FS demands (fs_salebook, fs_purcbook, fs_pournals).
- Build Logistics from Scratch (Q4 2025 – Q2 2026):** Front-end operational modules (Air/Ocean, Cartage, Vehicles/VIN) are built independently. Operations personnel encode logistics milestones and capture charges on a standardized Dual-Ledger (Billing Tab & Cost Tab).
- The Accounting Bridge Layer (Q3 2026):** Middleware automatically maps completed logistics charges into FS Sales/Purchase books upon file closure.
- Batch Simulation & GL Commitment (Q4 2026):** The accountant runs a "Trial Post" simulation (Debits == Credits). Once balanced, it commits to the General Ledger with real-time updates to the Balance Sheet and Income Statement.

2. STANDARD PROCESS FLOWCHART (LOGISTICS BUILD TO FS INTEGRATION)



3. STEP-BY-STEP IMPLEMENTATION ROADMAP (DELIVERY BY DEC 2026)

The table below illustrates how the engineering work is phased to guarantee zero disruption to the existing FS system:

Phase	Deliverables	Engineering Activities	FS Integration Impact
Phase 1 Q4 2025	Schema & Contract Definition	- Freeze the working FS database schema contract. - Map Logisuite charge codes (Ocean/Air/Cartage) to existing FS GL accounts. - Design intermediate staging tables for operational charges.	Zero risk. Existing working FS continues operating in production untouched.
Phase 2 Q1–Q2 2026	Logistics Construction (From Scratch)	- Build Ocean Export/Import modules (FCL/LCL, HBL/MBL). - Build Air Export/Import modules (IATA, AWB, flight manifests). - Build P/D Cartage & Vehicle Inventory (VIN decoder, title clearance). - Implement Dual-Ledger charge capture (Billing and Cost tabs).	Logistics operates independently; operational data accumulates in staging tables.
Phase 3 Q3 2026	Accounting Bridge Middleware	- Build the automated ETL translation service. - Implement the File Analysis Pre-Audit Gate (locking completed shipments). - Build automated Sales Book and Purchase Book voucher drafting. - Build Cash Disbursement Voucher (CDV) check generator.	Connects the newly built logistics staging tables to existing FS draft tables.
Phase 4 Q4 2026	End-to-End Testing & Go-Live	- Execute parallel test runs comparing Logisuite files against automated FS postings. - Validate Trial Post balance checks (Debits == Credits). - Final user training and production cutover before December 2026.	Full unified operation: operational shipment creation results in real-time FS statements.

4. DATA INTEGRATION CONTRACT (HOW DATA MOVES TO FS)

To guarantee the existing FS engine functions without modification, the new Logistics application writes to an interface mirroring the FS schema:

Logistics Source Field	Bridge Mapping Rule	Existing FS Destination	Ledger Effect
Billing Tab: Freight Charge	Map Client to AR Code; Amount to Revenue	fs_salebook → Trade AR	Debit: AR Trade Credit: Ocean/Air Revenue
Billing Tab: Fuel / Security / Doc Fees	Map line item code to ancillary GL	fs_salebook → Surcharges	Credit: Ancillary Freight Income
Cost Tab: Ocean/Air Line Carrier Bill	Map Carrier to Vendor AP Code	fs_purcbook → Trade AP	Debit: Cost of Goods Sold (COGS) Credit: AP Trade
Cost Tab: Port Charges / Duty Advances	Generate Disbursement Request	fs_purcbook / CDV Entry	Debit: Port Expense / Prepaid Duty Credit: Cash in Bank / AP

5. PROTECTION OF EXISTING WORKING FS ENGINE

To ensure developing the new Logistics engine does not destabilize the existing, working FS application, three engineering safeguards are enforced:

- **Physical Codebase Isolation:** All new logistics modules are built strictly inside the dedicated `kornet Express\kornet-system` repository. The production FS application (`web-system`) remains 100% pristine and unmodified.
- **Staging Buffer:** Operational staff cannot write directly to the General Ledger (`fs_pournals`). All logistics data lands first in a staging buffer. Only the Accounting Bridge, operated by an authorized accountant, can commit batches.
- **Validation & Rollback:** If a batch fails the Trial Post simulation (e.g., unbalanced debit/credit, inactive account code, or closed fiscal month), the bridge rejects the transfer and generates an audit discrepancy report without affecting existing GL balances.

Summary for Management: We do not rebuild or alter the working Financial Statements system. We use the working FS as our standard specification, build the unbuilt logistics modules around that standard, and link them using a strict, automated Accounting Bridge. This guarantees low project risk, predictable delivery before December 2026, and immediate operational continuity.